t: current time

T: closing time

S_t: underlying price at time t

Z_t: Geometric Brownian motion with volatility σ

Let $y \sim N(0,1)$:

$$(Z_T - Z_t) \sim N(0, T-t) = \sqrt{T-t}\, y$$

Derived from Ito's Lemma:

$$S_T = S_t e^{(\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\, y}$$

log return $\quad \ln\left(\frac{S_T}{S_t}\right) = (\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\, y$

log return norm $\quad \frac{\ln\left(\frac{S_T}{S_t}\right)}{\sigma\sqrt{T-t}} = \frac{\mu\sqrt{T-t}}{\sigma} - \frac{1}{2}\sigma\sqrt{T-t} + y$

Baseline: μ risk-free interest rate (around 4%)

$T-t$ time to maturity in year

σ atm IV

$$\boxed{N\left(\frac{\mu\sqrt{ttm}}{IV} - \frac{1}{2} IV\sqrt{ttm},\ 1\right)}$$